

The background features several decorative elements: a large solid blue circle in the top-left corner, a smaller circle with blue diagonal stripes in the top-right corner, another solid blue circle on the right edge, and a circle with blue diagonal stripes in the bottom-left corner. At the bottom, there are large, overlapping wavy shapes in shades of light blue and medium blue.

SALES ANALYSIS USING HYPOTHESIS TESTING

Technology vs Furniture
Sales Comparison

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APEXPLANET INTERNSHIP TASK 4

BUSINESS PROBLEM

The company wants to determine whether Technology products generate significantly different sales compared to Furniture products.

This analysis helps management make informed decisions regarding inventory planning, marketing investments, and resource allocation.

DATASET OVERVIEW



Dataset: Superstore Sales
Dataset

Total Records: 9800

Total Columns: 18

Key Variables:

1

Category

4

Customer Segment

2

Sub-Category

5

Sales

3

Region

HYPOTHESIS FORMULATION

NULL HYPOTHESIS (H_0)

AVERAGE SALES OF
TECHNOLOGY AND
FURNITURE
CATEGORIES ARE
EQUAL.

ALTERNATIVE HYPOTHESIS (H_1)

AVERAGE SALES OF
TECHNOLOGY AND
FURNITURE
CATEGORIES ARE
DIFFERENT.

PYTHON IMPLEMENTATION

```
hypothesis_test.py U X
Task 4 > hypothesis_test.py > ...
1 import pandas as pd
2 from scipy.stats import ttest_ind
3
4 df = pd.read_csv(
5     r"C:\Users\ASUS\Desktop\ApexPlanet-Internship\Task 3\train.csv"
6 )
7
8 technology = df[df["Category"] == "Technology"]["Sales"]
9 furniture = df[df["Category"] == "Furniture"]["Sales"]
10
11 t_stat, p_value = ttest_ind(
12     technology,
13     furniture,
14     equal_var=False
15 )
16
17 print("T-Statistic =", t_stat)
18 print("P-Value =", p_value)
```

Libraries Used:

- Pandas
- SciPy

Method:

Independent Two-Sample T-Test

HYPOTHESIS TEST RESULTS

**T-Statistic =
3.718**

**P-Value =
0.000205**

**Decision:
Reject H_0**

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ASUS\Desktop\ApexPlanet-Internship> & 'C:\Users\ASUS\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\ASUS\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '51693' '--' 'C:\Users\ASUS\Desktop\ApexPlanet-Internship\Task 4\hypothesis_test.py'
```

```
T-Statistic = 3.717907531778913
```

```
P-Value = 0.00020540281499218993
```

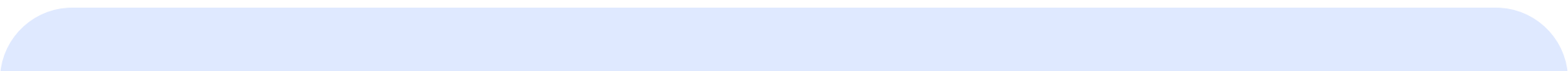
INTERPRETATION OF RESULTS



The p-value is significantly lower than 0.05.

This indicates a statistically significant difference between Technology and Furniture sales.

Therefore, category type has a measurable impact on sales performance.





BUSINESS INSIGHTS

01

Technology products generate stronger sales performance.

02

Product category influences revenue outcomes.

03

Marketing and inventory efforts can be optimized using these findings.

04

Data-driven decision making improves business efficiency.



CONCLUSION

- 1 Dataset analyzed successfully
- 2 T-Test performed using Python
- 3 Significant difference detected
- 4 Technology category outperformed Furniture
- 5 Business objective achieved

**THANK
YOU!**

QUESTIONS?

